

Code-Layout, Readability and Reusability

Date	28 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S. No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Followed PEP8 4-space indentation across app.py and HumDevIndex.ipynb
2	Proper File Structure	Files and folders are logically organized	Yes	Organized into Dataset, Flask (with templates), and Training folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Used names like x_train, y_pred, features_value, input_features
4	Function / Method Names	Functions are descriptively named	Yes	Flask routes named home(), prediction(), predict() matching purpose
5	Code Comments	Inline and block comments explain logic	Yes	Each notebook cell and app.py section has explanatory comments
6	Modular Design	Code is split into reusable functions/modules	Partial	Flask routes are modular; ML pipeline could be split further
7	No Redundant Code	Duplicate or unused code is removed	Yes	Removed unused imports and duplicate preprocessing cells
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation in predict() route; try-except can be expanded

Reusable Components / Modules:

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Pipeline	Python / Pandas, Scikit-learn	Used in HumDevIndex.ipynb for null handling and feature selection	High
2	HDI.pkl Model Loader	Python / Pickle	Loaded once at Flask app startup to serve all prediction requests	High
3	HTML Base Layout (Header/Footer)	HTML / CSS	Reused across home.html, indexnew.html, resultnew.html, dashboard.html	High
4	HDI Classification Logic	Python / Flask	if-elif block in predict() route classifying HDI categories	Medium

5	Chart.js Visualization Config	JavaScript / Chart.js	Reused chart styling across bar, line, pie and radar charts	Medium
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Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear separation of Dataset, Flask, and Training folders following ML project conventions
Readability	4	Descriptive variable names and inline comments make the code easy to follow
Reusability	4	Model loading, preprocessing pipeline and HTML layout components reused effectively
Documentation / Comments	4	Each notebook cell is commented; documentation covers setup and usage
Overall Score	4	Well-structured ML + Flask project with room for modularization improvements